

6610110554 นายปรเมษ แก้วอุบล

รายงานประสิทธิภาพโมเดล PhayaThaiBERT บน AMD GPU/ROCM

Model

- Base model: `clicknext/phayathaibert`
- Architecture: `AutoModelForSequenceClassification`
- Task: multi-label classification
- Activation: sigmoid
- Loss: BCEWithLogitsLoss
- Learning rate: `2e-05`
- Epochs: `15`
- Batch size: `8`
- Max length: `256`
- Device during train: `cuda`
- ROCm/HIP version: `6.2.41133-dd7f95766`

หมายเหตุ: `test.csv` ไม่มีเลข `tag` จึงวัด metric จาก validation split ของ `train.csv` และสร้าง submission จาก `test.csv`

```

parametk@UbuntuParametk:~/Desktop/AIforSocial/AIforSocial/W102$ .venv/bin/python train_bert.py --epochs 15
device: cuda
torch version: 2.5.1+rocm6.2
rocm/hip version: 6.2.41133-dd7f95766
HSA_OVERRIDE_GFX_VERSION: 11.0.0
gpu: AMD Radeon RX 7600S
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100% | 197/197 [00:00<00:00, 12585.72it/s]
[transformers] CamembertForSequenceClassification LOAD REPORT from: clicknext/phayathaibert
Key | Status |
-----+-----+
ln_head.dense.weight | UNEXPECTED |
ln_head.bias | UNEXPECTED |
ln_head.layer_norm.weight | UNEXPECTED |
ln_head.dense.bias | UNEXPECTED |
ln_head.layer_norm.bias | UNEXPECTED |
classifier.dense.weight | MISSING |
classifier.out_proj.bias | MISSING |
classifier.dense.bias | MISSING |
classifier.out_proj.weight | MISSING |

Notes:
- UNEXPECTED: can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING: those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
/home/parametk/Desktop/AIforSocial/AIforSocial/W102/.venv/lib/python3.12/site-packages/torch/nn/modules/linear.py:125: UserWarning: Attempting to use hipBLASLt on an unsupported architecture
! Overriding blas backend to hipblas (Triggered internally at ../aten/src/Aten/Context.cpp:296.)
  return F.linear(input, self.weight, self.bias)
/home/parametk/Desktop/AIforSocial/AIforSocial/W102/.venv/lib/python3.12/site-packages/transformers/integrations/sdpa_attention.py:92: UserWarning: Memory Efficient attention on Navi31 GPU is
still experimental. Enable it with TORCH_ROCM_AOTRITON_ENABLE_EXPERIMENTAL=1. (Triggered internally at ../aten/src/Aten/native/transformers/hip/sdp_utils.cpp:269.)
  attn_output = torch.nn.functional.scaled_dot_product_attention(
epoch 1/15 - train_loss=0.5089 valid_loss=0.3298
epoch 2/15 - train_loss=0.3016 valid_loss=0.2819
epoch 3/15 - train_loss=0.2790 valid_loss=0.2708
epoch 4/15 - train_loss=0.2562 valid_loss=0.2284
epoch 5/15 - train_loss=0.2172 valid_loss=0.1997
epoch 6/15 - train_loss=0.1843 valid_loss=0.1825
epoch 7/15 - train_loss=0.1539 valid_loss=0.1687
epoch 8/15 - train_loss=0.1323 valid_loss=0.1578
epoch 9/15 - train_loss=0.1118 valid_loss=0.1522
epoch 10/15 - train_loss=0.0998 valid_loss=0.1480
epoch 11/15 - train_loss=0.0903 valid_loss=0.1446
epoch 12/15 - train_loss=0.0830 valid_loss=0.1422
epoch 13/15 - train_loss=0.0769 valid_loss=0.1423
epoch 14/15 - train_loss=0.0751 valid_loss=0.1432
epoch 15/15 - train_loss=0.0727 valid_loss=0.1424
Writing model shards: 100% | 1/1 [00:00<00:00, 1.13it/s]
saved model: /home/parametk/Desktop/AIforSocial/AIforSocial/W102/bert_rocm_model

```

```
parametk@UbuntuParametK: ~/Desktop/AIforSocial/AIforSocial/W1D2 $ .venv/bin/python evaluate_bert.py  
device: cuda  
rocm/hip version: 6.2.41133-dd7f95766  
HSA_OVERRIDE_GFX_VERSION: 11.0.0  
gpu: AMD Radeon RX 7600S  
Loading weights: 100% [██████████] 201/201 [00:00<00:00, 12597.58it/s]  
/home/parametk/Desktop/AIforSocial/AIforSocial/W1D2/.venv/lib/python3.12/site-packages/torch/nn/modules/linear.py:125: UserWarning: Attempting to use hipBLASLt on an unsupported architecture! Overriding blas backend to hipblas (Triggered internally at ../aten/src/Aten/Context.cpp:296.)  
    return F.linear(input, self.weight, self.bias)  
/home/parametk/Desktop/AIforSocial/AIforSocial/W1D2/.venv/lib/python3.12/site-packages/transformers/integrations/sdpa_attention.py:92: UserWarning: Memory Efficient attention on Navi31 GPU is still experimental. Enable it with TORCH_ROCM_AOTRITON_ENABLE_EXPERIMENTAL=1. (Triggered internally at ../aten/src/Aten/nativeTransformers/hip/sdp_utils.cpp:269.)  
    attn_output = torch.nn.functional.scaled_dot_product_attention(  
saved submission: /home/parametk/Desktop/AIforSocial/AIforSocial/W1D2/Data/submission_bert_rocm.csv  
saved report: /home/parametk/Desktop/AIforSocial/AIforSocial/W1D2/bert_rocm_performance_report.md  
exact-match accuracy: 0.7383  
micro f1: 0.7815  
macro f1: 0.8078
```

Validation Metrics

- Exact-match accuracy: 0.7383
- Micro precision: 0.7686
- Micro recall: 0.7949
- Micro F1-score: 0.7815
- Macro precision: 0.8409
- Macro recall: 0.8020
- Macro F1-score: 0.8078
- Samples precision: 0.7664
- Samples recall: 0.7804
- Samples F1-score: 0.7695
- Hamming loss: 0.0405
- Samples Jaccard score: 0.7617